4. Skewness Calculation

a. data1 = [12, 15, 14, 10, 18, 20, 22, 25, 13, 17, 19, 21, 16, 14, 23]

Sol^n

Ascending order

10, 12, 13, 14, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 25

$N = 15$

Mean $(\bar{x}) = \frac{10+12+13+14+14+15+16+17+18+19+20+21+22+23+25}{15}$

$= \frac{259}{15} = 17.26$

Median, $\frac{N+1}{2} = \frac{15+1}{2} = \frac{16}{2} = 8$

8^{th} value = 17

$\therefore$ Median = 17

Now,

Standard Deviation

x	$x - \bar{x}$	$(x-\bar{x})^2$
10	-7.27	52.85
12	-5.27	27.77
13	-4.27	18.23
14	-3.27	10.69
14	-3.27	10.69
15	-2.27	5.15
16	-1.27	1.61
17	-0.27	0.07
18	0.73	0.53

19	1.73	2.99
20	2.73	7.45
21	3.73	13.91
22	4.73	22.37
23	5.73	32.83
25	7.73	59.75
Total		266.93

Now, Variance $= \frac{266.93}{15} = \frac{SD}{N}$

$= 17.80$

Standard Deviation $= \sqrt{variance}$

$= \sqrt{17.80}$

$= 4.22$

Skewness $= \frac{(mean - mode)}{SD}$

~~3(17.26~~

Using Pearson's second coefficient

$Sk = \frac{3(mean - median)}{SD}$

$= \frac{3(17.27 - 17)}{4.22}$

$= \frac{0.81}{4.22}$

$= 0.19$

b. data 2 = [5, 7, 8, 9, 10, 12, 15 18, 22, 30, 45, 60]

$N = 12$

$$\text{Mean}(\bar{x}) = \frac{5+7+8+9+10+12+15+18+22+30+45+60}{12}$$

$$= \frac{241}{12} = 20.08$$

Median

Since $n = 12$, 6^{th} & 7^{th} position

$$= \frac{12+15}{2} = 13.5$$

Now, Standard deviation

x	$x - \bar{x}$	$(x-\bar{x})^2$
5	-15.08	227.41
7	-13.08	171.09
8	-12.08	146.01
9	-11.08	122.76
10	-10.08	101.61
12	-8.08	65.29
15	-5.08	25.81
18	-2.08	4.33
22	1.92	3.69
30	9.92	98.41
45	24.92	621.01
60	39.92	1593.61
$\Sigma(x-\bar{x})^2$		3181.03

$$\text{Variance} = \frac{\sum (x-\bar{x})^2}{n}$$

$$= \frac{3181.03}{12}$$

$$= 265.09$$

$$\text{Standard deviation} = \sqrt{\text{variance}}$$

$$= \sqrt{265.09}$$

$$= 16.28$$

$$\text{Skewness (SK)} = \frac{3(\text{mean} - \text{median})}{SD}$$

$$= \frac{3(20.08 - 13.5)}{16.28}$$

$$= \frac{19.74}{16.28}$$

$$= 1.21$$